Schaeffer's

SPECIALIZED LUBRICANTS

ISO 9001 CERTIFIED

1. Substance/Product Identification

Product Name: 227 Moly Roller Chain Lube Spray
Company/Address: Schaeffer Mfg
102 Barton Street
Saint Louis, Missouri 63104
USA
Synonyms: None
Preparation/Revision Date: 05/09/2014
Product Use/Type: Chain lube and penetrating oil
Emergency Phone Number: +1 314 865-4105 (24-hour response number)
+ 1 314 865-4100 (Business hours 8:30AM-5:00PM)
1-800-325-9962 (US & Canada)
Website: www.schaefferoil.com
MSDS Number: 227Spray Version 1.0

2. Hazards Identification

Appearance: Black color
Odor: Solvent odor
Principal Hazards: Flammable Aerosol. Irritation of skin, eyes and upper respiratory tract

Hazardous Materials Information System (U.S.A)	Health: 2 Fire:4 Reactivity: 0
National Fire Protection Agency System (U.S.A.)	Health: 2 Fire: 4 Reactivity: 0

See Section 11 for complete health hazard information

Product Name: 227 Moly Roller Chain Spray

3. Composition and Information on Ingredients

Hazardous Ingredients

Ingredient Name	CAS No.	EU Number	Percentage (by wt.)
Petroleum distillates, hydrotreated heavy naphthenic	64742-52-5	265-155-0	30-60
Petroleum distillates, hydrotreated light naphthenic	64742-53-6	265-156-6	10-30
Odorless Minieral Spirits	8052-41-3	232-489-3	1-10
Graphite	7782-42-5	231-953-3	0.1-1.0
Molybdenum Disulfide	1317-33-5	215-263-9	1-5
Antimony Dialkyldithiocarbamate	15890-25-2	240-028-2	1-5
Propane (propellant)	74-98-6	200-827-9	1-5
Isobutane (propellant)	75-28-5	200-857-2	0.1-1.0

4. First Aid Measures

Ingestion: Do not induce vomiting. Give one or two glasses of water to drink. Never give anything by mouth to an unconscious person. Seek medical attention
Eyes: Immediately flush with clean water for 15 minutes. If irritation, redness or itching persists seek medical attention.
Inhalation: Remove victim to fresh air. If breathing is difficult administer oxygen by trained personnel. If upper respiratory tract irritation, headache, nausea, or drowsiness occurs If upper respiratory tract irritation, headache, nausea, or drowsiness occurs remove to fresh air. Get immediate medical attention if breathing becomes difficult or symptoms persist.
Skin: Remove contaminated clothing and launder before reuse. Wash skin with large amounts of soap and water. Seek medical attention if irritation develops
Additional Information: Aspiration into lungs may result in chemical pneumonitis. Dermatitis may result from prolonged or repeated exposure. High pressure injection injuries require prompt surgical intervention and possibly steroid therapy to minimize tissue damage and loss of function.

5. Fire Fighting Measures:

Flash Point: >140°C (>248°F) ASTM D-93 Pensky Martan Closed Cup Method
>180°C (>356°F) ASTM D-92 Cleveland Open Cup Method
Extinguishing Media: Dry chemical foam, water-fog, carbon dioxide
Firefighting Procedures: As in any fire wear a self contained breathing apparatus pressure demand, MSHA/NIOSH (Approved or Equivalent) and full protective gear
Unusual Fire & Explosion: Dense black smoke
Hazards: Hazardous combustion products may include a complex mixture of airborne solids and liquid particulates and gases, Carbon monoxide and unidentified organic and inorganic compounds

Product Name: 227 Moly Roller Chain Spray

6. Accidental Release Measures

Spill Procedures and Clean up methods: Use absorbent materials to contain spill. Collect used absorbent material and discard as dictated by Federal, State, Provincial, regional, national and local laws and regulations. Keep out of sewers and watercourses
Personal Precautions: Wear appropriate personal protective equipment when cleaning up a spill

Environmental Precautions: This material is not expected to present any environmental problems other than those associated with oil spills if spilled into a watercourse or waterway. If spilled in United States navigable waters the spill must be reported to the National Response Center (U.S. Coast Guard) or the appropriate governmental authorities if the product is spilled into rivers, creeks, streams or other natural waterways or courses. The National Response Center's Toll Free No. is 800-424-8802. If spilled into international or national waters report spill in accordance with all relevant international, national, provincial and local laws and regulations.

7. Handling & Storage

Handling: Avoid all contact of product with eyes and skin. Wash thoroughly after handling and before eating, smoking or using toilet facilities. Do not swallow product. Do not inhale vapors or mists. Do not eat, drink or smoke in work areas.

Storage: Do not store product at elevated temperatures. Store in a cool dry place that has adequate ventilation. Do not store near food or feed products. Keep containers closed when not in use.

8. Exposure Controls and Personal Protection

Occupational Exposure Limits

Ingredient	OSHA TWA	OSHA STEL	ACGIH TWA	ACGIH STEL
Complex mixture of petroleum base oils	5 mg/m^3		5 mg/m^3	
1-Decene Homopolymer Hydrogenated	5 mg/m^3		5 mg/m^3	
Graphite	15 mg/m^3 (Total dust)		3 mg/m^3 (Respirable fraction)	
Molybdenum Disulfide	15 mg/m^3 (total dust)		3.5 mg/m^3	
Dialkylpolysulfide	N/E		N/E	
Amine phosphate	N/E		N/E	
Polyisobutylene	N/E		N/E	
Carbon Black	3.5 mg/m^3		3.5 mg/m^3	
Organophilic Clay	NE		NE	

(s)-Skin exposure (p) proposed limit (c) – Ceiling limit (I) Recommended exposure limit
(u) supplier limit (N/E) – not established

Other Exposure Limits

Engineering Controls: Good general ventilation should be used. If applicable use process enclosures, local exhaust ventilation and other engineering controls to maintain airborne levels below recommended exposure limits.

Personal Protective Equipment

Gloves: Impervious oil resistant gloves such as Neoprene, PVC, Nitrile rubber
Protective Clothing: Apron and long sleeves recommended
Eye Protection: Chemical goggles or safety goggles
Respiratory Protection: None required under conditions of normal use

Product Name: 227 Moly Roller Chain Spray

9. Physical and Chemical Properties

Flash Point: >140°C (>248°F) ASTM D-93 Pensky Martan Closed Cup Method
>180°C (>356°F) ASTM D-92 Cleveland Open Cup Method
Specific Gravity: 1.03 – 1.08
Upper Flammable Limit (UEL): 1 (based on mineral oil)
Lower Flammable Limit (LEL): 10% (based on mineral oil)

Vapor Pressure: Not determined
Vapor Density (air =1): >1 (estimated value)
Percent Volatile: Nil
Evaporation Rate: Not determined
pH: Not applicable
Boiling Point: Base Oils >600°F (315°C)
Melting Point: >500°F (260°F) (dropping point)
Appearance: Black color
Odor: Slight hydrocarbon odor
Solubility in Water: Insoluble

The above data are typical values and do not constitute a specification. Vapor pressure data are calculated unless otherwise specified

10. Stability and Reactivity

Stability: Stable under normal conditions
Materials to Avoid: Strong oxidizing agents
Polymerization: Will not occur
Thermal Decomposition: Carbon dioxide and carbon monoxide. Byproducts of incomplete combustion
Conditions to Avoid: Avoid breathing mists and vapors

11. Toxicological Information:

Routes of Exposure: Skin, eyes, ingestion and inhalation
Target Organs: Skin, eye, upper respiratory tract

Acute Exposure

Ingestion: Nausea, diarrhea
Eye Contact: May cause irritation and redness to eyes
Skin Contact: Prolonged and repeated contact may cause irritation, redness and dryness to skin
Inhalation: Heating can generate vapors and mists that can cause nausea, dizziness and other upper respiratory irritation if inhaled.
Dermal Sensitization: Product is not considered to be a dermal sensitizer
Inhalation Sensitization: Product is not considered to be an inhalation sensitizer
Carcinogenicity: Product does not contain any components listed by ACGIH, IRAC, NIOSH, NTP or OSHA

Additional Toxicological Information: No data available to indicate the product or any of its components are present at greater than 0.1% are mutagenic, genotoxic or teatogenic. Used product may contain harmful impurities that have accumulated during use. The concentration of such harmful impurities will depend on use and they may present risks to health and the environment on disposal. All used grease should be handled with caution and skin contact avoided as far as possible.

Product Name: 227 Moly Roller Chain Spray

11. Toxicological Information Continued

Acute toxicity

Petroleum base oils
Acute Inhalation Toxicity: LC50 >5,000 mg/m3 (rat)
Acute Oral Toxicity LD50: >2,000 mg/kg (rat)

Acute Dermal Toxicity: LD50: >2,000 mg/kg (rabbit)

1-Decene Homopolymer, Hydrogenated
Acute Oral Toxicity: LD50 / rat / > 5000 mg/kg / based on data for a similar material.
Acute Dermal Toxicity: LD50 / rat / > 2000 mg/kg
Acute Inhalation Toxicity: LC50 / not known / based on data for a similar material

Organophilic Clay
Acute Oral Toxicity: LD50 / rat / > 8,000 mg/kg
Acute Dermal Toxicity:
Acute Inhalation Toxicity:

Polyisobutylene
Acute Oral Toxicity: LD50 / rat / > 34,600 mg/kg
Acute Dermal Toxicity: LD50/rat/>10,250 mg/kg
Acute Inhalation Toxicity: LC50/rat/>4,280 mg/m^3 (4 hours)

12. Environmental and Ecological Information

Ecotoxicity data has not been determined specifically for this product. Information is given based on a knowledge of the components and the ecotoxicology of similar products.

Ecotoxicity: Poorly soluble mixture, May cause physical fouling of aquatic organisms. Expected to be practically non toxic: LL/EL/IL50 >100 mg/l (to aquatic organisms) (LL/EL50 expressed as the nominal amount of product required to prepare aqueous test extract) Mineral oil, and 1-Decene Homopolymer, Hydrogenated are not expected to cause any chronic effects to aquatic organisms less than 1 mg/l.

Mobility: Product is semi-solid under most environmental conditions and floats on water. If it enters soil, it will absorb to soil particles and will not be mobile

Environmental Fate

Biodegradation: Expected not to be readily biodegradable
Bioaccumulation: Product has the potential to bioaccumulate, however metabolism or physical properties may reduce the bioconcentration or limit bioavailability.
Water Class Hazard (Germany): WK-2 according to appendix 4 VwVwS dated 27.7.2005
Other Adverse Effects: Product is a mixture of non-volatile components, which are not expected to be released to air in any significant quantities. Not expected to have ozone creation potential or global warming potential.

Product Name: 227 Moly Roller Chain Spray

13. Disposal Considerations

Waste Handling and Disposal:

Waste Handling and Disposal: This product in its neat state when discarded or disposed of is not a hazardous waste according to Federal Regulations 40CFR 261.4 (b)(4). Dispose of product in accordance with all applicable Federal, State and local laws and regulations. Do not re-use empty

containers dispose of in accordance with prevailing regulations. EU Waste Disposal Code (EWC): 12 01 12 spent waxes and fats. Classification of waste is always the responsibility of the end user

Container Warnings: Container is not designed to contain pressure. Do not use pressure to empty container or it may rupture with explosive force. Empty containers retain product residue (solid, liquid and/or vapor) and can be dangerous. Do not cut, weld, braze, solder, drill, grind or expose containers to heat, flame, sparks, static electricity or other sources of ignition. There may explode and cause injury or death. Empty containers should be completely drained, properly closed and promptly returned to a drum reconditioner or disposed of properly

14. Transport Information

US DOT Classification Non-Bulk: Not regulated as a hazardous material or dangerous good for transportation.

US DOT Classification Bulk: Steel Drum, UN1993, Flammable Liquids, N.O.S., (Heavy Alkylated Petroleum Naphtha, Acetone),3, PgIII

IMDG Classification:
Non-Bulk: Fibreboard Box Stc 12 X 16 Oz Aerosol Cans (5443 Grams), UN1950, Aerosols, 2.1, (Flash Point 38.C Pmcc), Limited Quantity
Bulk: Steel Drum, UN1993, Flammable Liquids, N.O.S., (Heavy Alkylated Petroleum Naphtha, Acetone),3, PgIII, (Flash Point 32c, Pmcc)

IATA/ICAO Classification: Not regulated as a hazardous material or dangerous good for transportation

ADR: Not regulated as a hazardous material or dangerous good for transportation.

ADNR: Not regulated as a hazardous material or dangerous good for transportation.

RID: Not regulated as a hazardous material or dangerous good for transportation.

Canadian TDG: Not regulated as a hazardous material or dangerous good for transportation.

15. Regulatory Information

Hazard Symbols: None

EC Classification: Not classified as dangerous under EC criteria
EC Risk Phrase: Not classified
R 43 May cause sensitization by skin contact
EC Safety Phrase: Not classified
EINECS: All components listed or polymer exempt

Product Name: 227 Moly Roller Chain Spray

15. Regulatory Information continued

GHS Symbol:
Health Hazards: May cause sensitization by skin contact

TSCA Inventory: All of the components in this material are on the US TSCA Inventory or are exempt.

State of California Proposition 65: This product does not contain any of the chemicals that are listed on the State of California's Proposition 65 List as potential cancer causing agents or reproductive toxins

US EPA SARA Title III and CERCLA Listings and Reportable Quantities
US EPA Section 311/313 Classifications
This product does not contain greater than 1% of any chemical substance listed under the SARA Title III Section 313, CERCLA's 302/304 and 102(a) Lists
Section 311/312 (40CFR 370)

Acute (Immediate)	Chronic (Delayed)	Fire	Reactive	Pressure Generating
X				

US Tariff Heading Number: 3403.19.0000
Schedule B Number: 3403.19.0000

Volatile Organic Compounds: None

Other Regulations:

Canada: All of the ingredients of this product are in compliance with the Canadian Environmental Protection Act and are present on the Domestic Substance List

Canadian WHMIS Classifications and Symbols: This product is not considered to be a WHMIS Hazard

EEC: All components are in compliance with the EC Seventh Amendment Directive 92/32/EEC

16. Other Information

For additional information call +1 314-865-4100 (outside the US and Canada) or 1-800-325-9962 inside the United States and Canada.

Although the information and recommendations set forth herein (hereafter referred to as information) are presented in good faith and believed to be accurate and factual as of the date hereof, Schaeffer Mfg. Company makes no representation as to the completeness or accuracy thereof. Information is supplied upon the condition that the person receiving the same will make their own determination as to its safety and suitability for their purposes prior to use. In no event will Schaeffer Mfg. Company be responsible for damages of any natures whatsoever resulting from the use or reliance upon information. ***No representation or warranty, either expressed or implied, of merchantability or fitness for a particular purpose is made with respect to information of the product to which the information refers. Compliance with all applicable federal, state, and local regulations remains the responsibility of the user.***